1. PROBLEM STATEMENT

More than 10,000 children die every day from hunger [1]. Farmers in the United States are unable to meet the domestic demands for food, with imports exceeding exports by over $41 billion [2]. Additionally, farmland is lacking in nutrients; at least one soil or water resource concern exists on 49% of surveyed soybean, oat, cotton, and wheat fields [3]. The process for testing soil for nutrients is long, requiring multiple individual samples to create a representative composite sample. Depending on the sampling methodology, extension services generally recommend collecting approximately 10–20 cores from each management area or zone [4]. Lab tests cost around $22-$30 [5], [6].

1.1. The Need

Current attempts to address this issue are flawed. The ROGO soil sampling robot follows a pre-programmed grid and collects soil samples to send off to another lab for testing. It is expensive to own and to operate, running on diesel [7]. The Agrobot Lala does in-field analysis, but it can only measure nitrate, takes about 30 minutes per sample, can only run for 3 hours on its battery, is more inaccurate than official lab results, and costs roughly €50,000 [8].

The best way to get accurate and complete measurements of the quality of soil is an official lab test, but testing every point on a plot of land is costly and time consuming, taking up to 15 days to get a result [5]. There is a need for a fast, low-cost way to identify which areas of a field vary enough to justify sending samples to a lab8

1.2. Our Solution

We propose a Digital Instrument for Remote Terrain Testing (D.I.R.T.T.), an autonomous, solar-powered robot that surveys a large plot and identifies areas where soil conditions differ enough to warrant laboratory testing. Instead of replacing the lab, D.I.R.T.T. tells where to send samples from, so that the costly lab analysis is spent where it is most needed. D.I.R.T.T. measures four soil properties, each chosen for a specific agronomic value: pH, electrical conductivity (EC), moisture, and temperature.

1.3. Technical Background

pH governs the availability of nutrients to plants, and it can vary wildly within a single plot of land. One mapping study measured a 2.9-unit range of pH in one field, and uniform lime application (which is the standard way to reduce acidity in soil) would leave over 63% of the land over-limed [9]. Mapping where pH differs shows where acidity problems are concentrated.

Electrical conductivity responds to several soil properties all at once, including clay content, moisture, salinity, cation exchange capacity, and organic content [10]. Because of this, EC maps are widely used to delineate management zones and to guide where soil samples are to be taken, giving a more representative set than traditional grid sampling [10].

Moisture and Temperature are used to gain more accurate EC readings [10], [11]. Moisture is useful for detecting drainage problems, one of the largest problems with soil health [12]. Temperature is useful for detecting biological activity[13].

D.R.I.T.T. is designed to address the cost and operating limitations of existing robots. Where the ROGO robot runs on diesel [7] and the Agrobot Lala costs roughly €50,000 [8], D.I.R.T.T. is solar powered, removing fuel costs and extending the time it can operate in the field.

As D.I.R.T.T. travels the plot, it logs each reading with its GPS location and plots the data of the field. Areas where readings fall outside expected ranges, or change sharply, are highlighted as candidates for lab sampling. The farmer can then collect a small number of samples from those areas, plus a few from areas that look normal to confirm the screening, instead of sampling the whole field.

References

[1] United Nations, "Losing 25,000 to hunger every day," UN Chronicle, vol. 45, no. 2–3, Jun. 2008. [Online]. Available: https://www.un.org/en/chronicle/article/losing-25000-hunger-every-day

[2] USDA Economic Research Service, "U.S. agricultural import values outpaced export values in 2025,"USDA ERS, Jun. 30, 2026. [Online]. Available: https://www.ers.usda.gov/data-products/chart-gallery/58310

[3] A. B. Rosenberg and S. Wallander, "Farmer-reported soil and water concerns vary by crop type and region," USDA Economic Research Service, Amber Waves, May 6, 2022. [Online]. Available: https://www.ers.usda.gov/amber-waves/2022/may/farmers-report-soil-related-resource-concerns-on-bout-half-of-soybean-wheat-cotton-and-oat-fields

[4] NC State Extension, "Soil sampling strategies for site-specific field management," NC State Extension Publications. [Online]. Available: https://content.ces.ncsu.edu/soil-sampling-strategies-for-site-specific-field-management

[5] University of Minnesota Soil Testing Laboratory, "Lawn & garden soil testing services and fees," Univ. of Minnesota, St. Paul, MN, USA, 2026. [Online]. Available: https://soiltest.cfans.umn.edu/testing-services/lawn-garden. Accessed: Oct. 1, 2026.

[6] UW–Madison Division of Extension, Door County, "Lawn and garden soil sampling," Sturgeon Bay, WI, USA, 2026. [Online]. Available: https://door.extension.wisc.edu/files/2026/07/LAWN-AND-GARDEN-SOIL-SAMPLING-instructions-1.pdf. Accessed: Oct. 1, 2026.

[7] B. Barker, "Autonomous soil sampling: Robo sampling improves accuracy, "Top Crop Manager, https://www.topcropmanager.com/autonomous-soil-sampling-robo-sampling-improves-accuracy/

[8] G. Kitić et al., "Agrobot Lala—an autonomous robotic system for real-time, in-field soil sampling, and analysis of nitrates," Sensors, vol. 22, no. 11, p. 4207, May 2022. Doi:10.3390/s22114207

[9] E. Bönecke et al., "Guidelines for precise lime management based on high-resolution soil pH, texture and SOM maps generated from proximal soil sensing data," *Precis. Agric.*, vol. 22, pp. 493–523, 2021. [Online]. Available: https://oa.tib.eu/renate/handle/123456789/6365. Accessed: Oct. 1, 2026.

[10] Sugar Research Australia, "Soil apparent electrical conductivity," Sugar Research Australia, Brisbane, QLD, Australia, Information Sheet IS14016, 2025. [Online]. Available: https://sugarresearch.com.au/wp-content/uploads/2025/04/IS14016-Soil-Apparent-Electrical-Conductivity.pdf. Accessed: Oct. 1, 2026.

[11] D. Mori, et al., "Calibration complexities: Full-scale error impact and simultaneous variation of salinity, temperature, and moisture content on sensor performance in soil," Indian Inst. Technol. Mandi, Mandi, India. [Online]. Available: https://scene.iitmandi.ac.in/pages/publication/publication_details/760. Accessed: Oct. 1, 2026.

[12] "Variable rates and your soil," *Grainews*. [Online]. Available: https://www.grainews.ca/machinery/variable-rates-and-your-soil. Accessed: Oct. 1, 2026.

[13] New Mexico State University Cooperative Extension Service, "Soil biological processes," Las Cruces, NM, USA, Guide A153. [Online]. Available: https://pubs.nmsu.edu/_a/A153. Accessed: Oct. 1, 2026.